Example: Find a Taylor series for $f(x) = \ln(x)$ about $x = 1$.

	$f^{(n)}(x)$	$f^{(n)}(1)$	
f	$\ln(x)$	0	
f'	$\frac{1}{x} = x^{-1}$	1	$= 0!$
f''	$-x^{-2}$	-1	$= -(1!)$
f'''	$2x^{-3}$	2	$= 2!$
f''''	$-6x^{-4}$	-6	$-(3!)$

$$= 0 + 1(x-1) - \underset{\frac{1}{2}}{\cancel{\frac{1}{2!}}}(x-1)^2 + \underset{\frac{1}{3}}{\cancel{\frac{2!}{3!}}}(x-1)^3 - \underset{\frac{1}{4}}{\cancel{\frac{3!}{4!}}}(x-1)^4 + \ldots$$

$$= 0 + 1(x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-3)^3 - \frac{1}{4}(x-1)^4$$

First term is 0 so start @ 1 or else divide by 0

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}(x-1)^n$$

Evens are the ones that are negative

Example: Consider $f(x) = \ln x$

(a) Find the Taylor series for $f(x) = \ln x$ at $a = 1$:

(from last page 6)

$$\ln(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(x-1)^n}{n}$$

(b) Find a Taylor polynomial of degree $n = 3$ at $a = 1$ to approximate $f(x)$.

First degree 2nd degree 3rd degree

$$T_3(x) = 0 + (x-1)^1 - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3$$

(c) Use Taylor's inequality to estimate the accuracy of the approximation $f(x) \approx T_3(x)$ for $0.9 \le x \le 1.1$. How well does $T_3(x)$ approximate $\ln(x)$?

↳ How big can $R_3(x) = \ln(x) - T_3(x)$ be, for $0.9 \le x \le 1.1$?

$|R_3(x)| \le \frac{M}{4!}|x-1|^4$ where M is bound for $f^{(4)}(x)$

- $f^{(4)}(x) = \frac{-6}{x^4}$, $|f^{(4)}(x)| = \frac{6}{|x|^4} \le \frac{6}{(0.9)^4} = M$ What are we doing here

So $|R_3(x)| \le \frac{6}{4!(0.9)^4}|x-1|^4$ Bounded at endpoints – local max occurs @ endpoints
↳ See prev. example

$$|R_3(x)| \le \frac{6}{4!(0.9)^4}|x-1|^4 \le \frac{6}{4!(0.9)^4}(0.1)^4 \approx \underbrace{0.000039}_{\text{Remainder}}$$

$R_3(x) = \ln(x) - T_3(x)$

$$|R_K(x)| \le \frac{M}{(K+1)!}|x-a|^{K+1}$$

$$R_K(x) = \sum_{n=K+1}^{\infty} \frac{f^{(n)}(c)}{(n!)}(x-a)^n$$

Example: Consider $f(x) = \ln(1 + 2x)$

(a) Find a Taylor polynomial of degree $n = 3$ at $a = 1$ to approximate $f(x)$.

1. Find Taylor series for $f(x)$ → Don't need to find series pattern

$f^{(n)}(x)$	$f^{(n)}(1)$
$f(x) = \ln(1+2x)$	$f(1) = \ln(3)$
$f'(x) = (1+2x)^{-1} \cdot 2$	$f'(1) = \frac{2}{3}$
$f''(x) = -(1+2x)^{-2} \cdot 2 \cdot 2$	$f''(1) = \frac{-4}{3^2}$
$f'''(x) = 2(1+2x)^{-3} \cdot 8$	$f'''(1) = \frac{16}{3^3}$
$f'''' = -6(1+2x)^{-4} \cdot 16$	$f''''(1) =$

We need a bound for something that's 1 degree more, though approx only goes to f'''; we need f'''' for a bound

Since we only need degree 3 it's fine

2 strategies for getting M (bound on derivative)
(1) $|\cos(x)|, |\sin(x)| \leq 1$ ← easy to get bound here since they're already bounded
(2) Fraction ex $\frac{5}{x^2}$ → Plug in smallest x

2. Approximation

$$T_3(x) = \ln(3) + \frac{2}{3}(x-1) - \frac{4}{3^2 \cdot 2!}(x-1)^2 + \frac{16}{3^3 \cdot 3!}(x-1)^3$$

(b) Use Taylor's inequality to estimate the accuracy of the approximation $f(x) \approx T_3(x)$ for $0.5 \leq x \leq 1.5$.

How big can $R_3(x)$ be?

$$|R_3(x)| \leq \frac{M}{4!}|x-1|^4 \quad \text{where } M \text{ bounds } f^{(4)}(x)$$

$$f^{(4)}(x) = \frac{-6 \cdot 16}{(1+2x)^4}$$

We want to make this as big as possible so denom must be as small as possible

$$\underbrace{|f^{(4)}(x)|}_{\text{abs value so always +}} \leq \frac{6 \cdot 16}{(1+2(0.5))^4} = \frac{6 \cdot \cancel{16}}{\cancel{2^4}} = 6 = M$$

$2^4 = 16$

$$|R_3(x)| \leq \frac{6}{4!}\underbrace{|x-1|^4}_{\text{local max at endpoints}} \leq \frac{6}{4!}(0.5)^4$$

$$\frac{6}{4!}(0.5)^4 \approx 0.016$$

Even if we haven't found Taylor series we can still find decent approx?

R = how much $T_k(x)$ is off from $f(x)$

Increasing degree makes R smaller

2 types of Diff Eqs:
1. Separable Eq.
2. Power Series solutions

Differential Equation: A *differential equation* is an equation that involves an unknown function $f(x)$ and its derivatives. Common examples include:

$$f'(x) = f(x), \qquad f''(x) = -f(x).$$

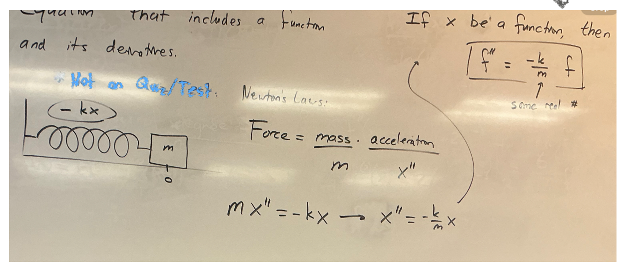
The *order* of the differential equation is given by the largest number of derivatives that appear.

↳ Number of derivatives that show up

Separable Differential Equations: A *separable equation* is a first-order differential equation in which

$$\frac{dy}{dx} = g(x)h(y)$$

for $y = f(x)$.



Example: Solve for $y = f(x)$ given

$y' = \frac{x^2}{y^2}$ → $$\frac{dy}{dx} = \frac{x^2}{y^2}.$$

↳ First order b/c only 1 derivative

$\frac{dy}{dx} = \frac{x^2}{y^2} \to y^2 \frac{dy}{dx} = x^2 \to y^2\,dy = x^2\,dx \to \int y^2\,dy = \int x^2\,dx$

bad Notationally?

$\frac{1}{3}y^3 = \frac{1}{3}x^3 + C$

you only need 1 C even if you integrate both sides

$y^3 = x^3 + 3C$

$y = (x^3 + 3C)^{\frac{1}{3}}$ C could be anything

To get a unique solution, need a boundary/initial conditions

↳ # of boundary conditions you need = order of diff eq

Ex: f(0) = 2

$2 = (0^3 + 3C)^{\frac{1}{3}}$

$2 = 3^{\frac{1}{3}} C^{\frac{1}{3}}$

$C^{\frac{1}{3}} = \frac{2}{3^{\frac{1}{3}}} \to C = \frac{2}{3}$

So here, we would need 1 boundary condition so order is 1

Note that to get a unique solution, we often have to specify additional information in the form of *boundary conditions*. If $f(0) = 1$, what is the solution to the example above?

Example: Find an expression for $y = f(x)$ with $f(0) = \pi$ given

$$\frac{dy}{dx} = \frac{6x^2}{2y + \cos(y)}.$$